

Role Prompting — Reference List

Use as: `Acting as a <role>, ...`

The role shifts Copilot's lens — what it prioritises, what it flags, what vocabulary it uses.

Engineering Roles

Role	What it shifts
Senior TypeScript / Java / Python / Go engineer	Language idioms, best practices for that ecosystem
Backend engineer	API design, data access patterns, server-side concerns
Frontend engineer	Component design, accessibility, bundle size awareness
Full-stack developer	End-to-end tradeoffs, API contract ownership
Systems programmer	Memory management, performance, low-level concerns
Embedded / firmware engineer	Resource constraints, determinism, no dynamic allocation
DevOps / Platform engineer	Infra-as-code, CI/CD, observability, deployment safety
Site Reliability Engineer (SRE)	Reliability, SLOs, error budgets, graceful degradation
Data engineer	Pipeline design, schema evolution, idempotency
ML engineer	Data preprocessing, model integration, inference efficiency

Architecture Roles

Role	What it shifts
Software architect	Design patterns, SOLID, long-term maintainability
Solution architect	System-level tradeoffs, component boundaries, vendor choice
Enterprise architect	Governance, standards compliance, org-level consistency
API designer	REST/GraphQL conventions, versioning, consumer-first thinking
Database architect	Normalisation, indexing strategy, query performance
Cloud architect (AWS / Azure / GCP)	Managed services, cost optimisation, cloud-native patterns
Microservices architect	Service boundaries, coupling, distributed systems tradeoffs
Event-driven architect	Messaging patterns, eventual consistency, idempotency

Security Roles

Role	What it shifts
------	----------------

Application security engineer	OWASP Top 10, input validation, secure defaults
Penetration tester	Attacker mindset — "how would I exploit this?"
Security auditor	Compliance evidence, audit trail, policy adherence
Cryptographer	Algorithm selection, key management, random number generation
Identity / IAM engineer	Auth flows, token handling, privilege escalation risks
Data privacy engineer	PII exposure, GDPR / DPDP compliance, data minimisation

Quality & Testing Roles

Role	What it shifts
Senior code reviewer	Readability, naming, abstraction leaks, PR comments
QA / Test engineer	Edge cases, boundary conditions, test coverage gaps
Performance engineer	Algorithmic complexity, memory allocation, profiling approach
Accessibility expert	WCAG compliance, screen reader compatibility, keyboard navigation
Localisation engineer	i18n patterns, string externalisation, locale-aware formatting

Data & Database Roles

Role	What it shifts
Database administrator (DBA)	Index strategy, locking, connection pooling, vacuum/maintenance
SQL expert	Query optimisation, execution plans, join strategy
Data modeller	Entity relationships, normal forms, schema evolution
Elasticsearch engineer	Mapping design, query DSL, relevance tuning
Redis architect	Data structure selection, eviction policy, persistence tradeoffs

Documentation & Communication Roles

Role	What it shifts
Technical writer	Clarity, audience-appropriate language, structure
API documentation specialist	OpenAPI conventions, examples, error descriptions
Developer advocate	Tutorial-style explanation, onboarding experience
Rubber duck debugger	Forces step-by-step reasoning aloud — great for debugging prompts

Domain Expert Roles (Industry)

Role	What it shifts
Financial systems engineer	Precision arithmetic, audit requirements, regulatory constraints
Payments engineer	PCI-DSS awareness, idempotency, double-spend prevention
Healthcare data engineer	HIPAA / HL7 / FHIR, PHI handling, consent management
E-commerce engineer	Inventory concurrency, cart consistency, tax calculation edge cases
Compliance engineer	Regulatory mapping, evidence collection, policy implementation

Teaching & Explanation Roles

Role	What it shifts
Senior engineer mentoring a junior	Explains the why, not just the what. Adds comments.
Computer science professor	Theory-first, complexity analysis, formal definitions
Code explainer	Line-by-line walkthrough, plain English, no jargon
Interviewer	Asks clarifying questions before answering — surfaces assumptions

Quick Examples

Acting as a senior SRE, review this deployment script for single points of failure.
Acting as a penetration tester, identify how this login endpoint could be abused.
Acting as a DBA, rewrite this query to avoid a full table scan.
Acting as a technical writer, rewrite this README for a developer new to the codebase.
Acting as a payments engineer, review this refund flow for idempotency issues.
Acting as a senior code reviewer, give me the 3 most important comments you would leave on this pull request.
Acting as a computer science professor, explain the time complexity of this algorithm and suggest a more efficient approach.

Tips for Role Prompting

- **Be specific.** "Senior TypeScript engineer" produces better output than "software engineer."
- **Combine roles** when needed: "Acting as a senior Java engineer with a focus on Spring Security..."
- **The role changes emphasis, not facts.** Copilot still uses the same model — the role shifts what it prioritises and flags.

- **Stack with constraints:** Role prompting + Constraint-based = very precise output.

Acting as a DBA reviewing for production safety.

CONSTRAINTS: flag anything that could cause a table lock, full scan, or N+1 query.